

MARINE SPILL INTELLIGENCE SYSTEM

An AI-Powered Satellite-Based Platform for Marine Oil Spill
Detection, Drift Analysis and Vessel Investigation

Problem Statement ID: SIH26143

Theme: Space Technology | Category: Software

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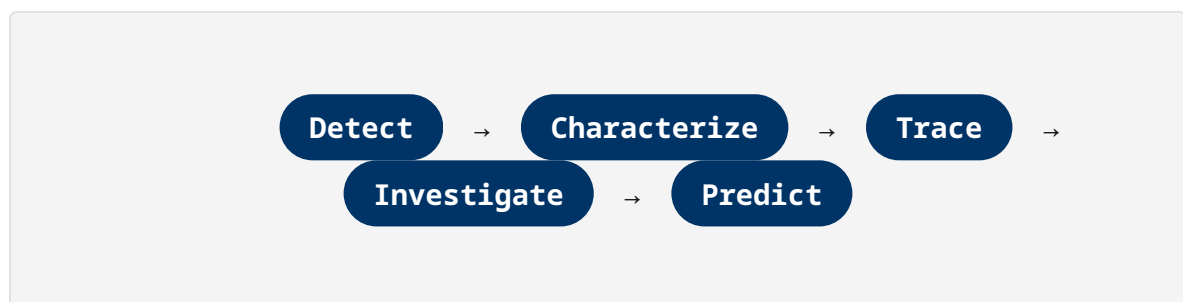
1. EXECUTIVE SUMMARY

Marine oil spills pose significant threats to marine ecosystems, coastal environments, fisheries, wildlife, and maritime economies. Effective response requires more than identifying the presence of an oil slick. Authorities must determine the extent and characteristics of the spill, estimate its possible origin, predict its future movement, and investigate vessel activity associated with the affected region.

The Marine Spill Intelligence System (MSIS) is proposed as an integrated geospatial intelligence platform that combines satellite imagery, artificial intelligence, environmental data, and Automatic Identification System (AIS) vessel data to support the end-to-end analysis of marine oil spills.

Core Functions:

- Detection of potential oil spills from satellite imagery
- Differentiation between oil spills and visually similar phenomena
- Segmentation and characterization of detected spill regions
- Estimation of likely spill origin through backward drift analysis
- Prediction of potential future movement through forward drift modelling
- Reconstruction and analysis of vessel traffic using AIS data
- Ranking of vessels for investigation based on spatio-temporal and behavioural evidence
- Visualization of results through an interactive geospatial interface



2. PROBLEM STATEMENT

Marine oil spills are difficult to monitor and investigate because the process involves multiple independent sources of information. Satellite imagery can identify potential oil slicks on the ocean surface. However, detection alone does not answer critical operational questions regarding exact location, extent, authenticity (vs. look-alikes), origin, future movement, and vessel accountability.

3. PROPOSED SOLUTION

MSIS integrates four primary intelligence components into a unified workflow:

1. **Satellite-Based Oil Spill Detection:** AI analysis of SAR imagery.
2. **Spill Characterization:** Geographic and physical segmentation.
3. **Drift Analysis:** Hindcasting (origin) and forecasting (future movement).
4. **Vessel Investigation:** AIS data reconstruction and ranking.

4. PROJECT OBJECTIVES

- **4.1 Oil Spill Detection:** Automatic detection using ML/CV.
- **4.2 Look-Alike Differentiation:** Reducing false positives.
- **4.3 Spill Segmentation:** Precise spatial boundary identification.
- **4.4 Spill Characterization:** Extracting location, area, geometry, and confidence.
- **4.5 Origin Estimation:** Backward drift modeling.
- **4.6 Movement Forecasting:** Environmental condition-based predictions.
- **4.7 Vessel Investigation:** Reconstructing vessel traffic.
- **4.8 Explainable Prioritization:** Transparent evidence-based ranking.

5. SYSTEM SCOPE

5.1 NEAR-REAL-TIME MONITORING

Select Region → Retrieve Latest Satellite Observation →
Preprocess → AI Detection → Spill Characterization →
Environmental Analysis → Vessel Investigation →
Interactive Visualization

5.2 HISTORICAL EVENT ANALYSIS

Select Historical Event → Load Satellite Data → Detect &
Segment → Characterize → Hindcasting → Origin Estimation
→ AIS Analysis → Priority Ranking → Forward Drift →
Visualization

5.3 UPLOAD AND ANALYZE

Supports **Georeferenced Satellite Data** (GeoTIFF) for full analysis and
Standard Image Data (PNG/JPG) for visual detection/segmentation only.

6. COMPLETE SYSTEM WORKFLOW

[DATA INPUT]





7-9. DETECTION, LOOK-ALIKES & SEGMENTATION

Utilizing **Sentinel-1 SAR** imagery (Synthetic Aperture Radar) which operates independently of light and cloud cover. The AI pipeline distinguishes true oil spills from "look-alikes" (e.g., natural slicks, wind patterns) and generates high-fidelity segmentation masks for area calculation.

10-14. DRIFT & VESSEL INVESTIGATION

Hindcasting: Backward simulation using ocean currents and wind to find the "Likely Origin Zone."

Forecasting: Predicting trajectory for the next 6, 12, and 24 hours.

AIS Investigation: Filtering vessels by spatial/temporal proximity and analyzing behavioral anomalies (e.g., AIS gaps or speed changes) to produce an **Investigation Priority Score**.

24. TECHNOLOGY STACK

Category	Technology
Languages	Python, TypeScript
AI/CV	PyTorch, OpenCV, NumPy, Scikit-learn
Geospatial	Rasterio, GeoPandas, Shapely, GDAL
Drift Modeling	OpenDrift, OpenOil
Backend	FastAPI
Frontend	React, MapLibre GL / Leaflet
Database	PostgreSQL + PostGIS

31. PROJECT VALUE PROPOSITION

MSIS transforms individual data points into an **integrated intelligence pipeline**. Instead of isolated detection, it provides a comprehensive Detect → Trace → Investigate workflow, offering actionable insights for maritime authorities.

32. CONCLUSION

The Marine Spill Intelligence System establishes a technically feasible, scalable framework for marine environmental protection and maritime law enforcement through the convergence of space technology and artificial intelligence.